Dividing with Remainders

Let's divide with remainders and decide what the remainder means.

Warm-up

Notice and Wonder: Expressions and Equations

What do you notice? What do you wonder?

division expression	multiplication equations
$100 \div 5$	$100 = 20 \times 5$
$101 \div 5$	$101 = 20 \times 5 + 1$
$102 \div 5$	$102 = 20 \times 5 + 2$
$103 \div 5$	$103 = 20 \times 5 + 3$
$104 \div 5$	$104 = 20 \times 5 + 4$
$105 \div 5$	$105 = 20 \times 5 + 5$

Activity 1

Is There a Remainder?

1. Decide if each expression will result in a remainder. Explain how you know.

   a. $753 \div 6$

   b. $918 \div 9$

   c. $1{,}263 \div 2$

   d. $2{,}630 \div 5$

2. Find the value of 2 expressions. Choose one with a remainder and one without. Explain or show your reasoning.

Activity 2

Hearts and Seats

1. Two artists paint 378 wooden hearts. They sell the hearts in boxes of 4.
   - The first artist says they will need 94 boxes for all the hearts.
   - The second artist says 95 boxes are needed.

   Who do you agree with? Explain or show your reasoning.



2. An auditorium seats 258 people. The seats are arranged in rows of 9, but there is one short row with fewer than 9 seats.

   How many rows of 9 seats are there? How many seats are in the shorter row?